MIL-C-21049E
16 September 1985
SUPERSEDING
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2 March 1977

MILITARY SPECIFICATION

COUPLING ASSEMBLIES, QUICK DISCONNECT, AIRCRAFT LIQUID OXYGEN SYSTEMS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of liquid oxygen quick-disconnect coupling assemblies and coupling halves.

* 1.2 Classification. Coupling assemblies shall be of the following types, as specified (see 6.2.1b):

Type I - Supply coupling assemblies

Type II - Vent coupling assemblies

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

O-T-620	1, 1, 1, Trichloroethane, Technical, Inhibited (Methyl Chloroform)
PPP-B-566	Box, Folding, Paperboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Systems Engineering and Standardization Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1660

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SPECIFICATIONS (Cont'd)

FEDERAL

PPP-B-636	Box, Shipping, Fiberboard
PPP-B-665	Box, Paperboard, Metal Edged and Components
PPP-B-676	Box, Setup
PPP-C-843	Cushioning, Material, Cellulosic

MILITARY

MIL-P-116	Preservation-Packaging, Methods of
MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-O-27210	Oxygen, Aviator's Breathing, Liquid and Gas
MIL-C-81302	Cleaning Compound, Solvent, Trichlorotrifluoroethane
MIL-T-81533	1, 1, 1 Trichloroethane (Methyl Chloroform), Inhibited, Vapor Degreasing

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-143	Standards and Specifications, Order of Precedence for The Selection of
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-889	Dissimilar Metals
MS22068	Coupling Assemblies, Quick Disconnect, Aircraft Liquid Oxygen Systems

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies of the above publication should be addressed to the Uniform Classification Committee, 202 Chicago Union Station, Chicago, IL 60606).

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* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Qualification. The couplings furnished under this specification shall be a product which is qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3, 4.3.1, and 6.3). In addition, the retention of the qualification for the couplings on the applicable qualified products list shall be dependent on periodic verification of continued compliance with the requirements of this specification (see 4.3 and 4.3.2).

* 3.2 First article. When specified, the coupling shall be subjected to first article inspection (see 4.4. and 6.4).

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials. Materials shall conform to applicable specifications and shall be as specified herein and on applicable drawings. Materials which are not covered by specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

* 3.4.1 Metal parts. All metal parts shall be of a corrosion-resistant material or treated in a manner to render them adequately resistant to corrosion.

3.4.1.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.4.2 Protective treatment. When materials are used in the construction of the couplings that are subject to deterioration when exposed to environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of environmental conditions shall be avoided.

3.5 Design and construction. The design and construction of the coupling shall be in accordance with MS22068. The outer sleeve of the female coupling half shall rotate freely about the fitting end and center body of the female coupling half, in the uncoupled condition.

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3.5.1 Type I coupling. The type I coupling shall contain a check valve in each coupling half. When the coupling is connected, liquid oxygen shall flow. When the coupling is disconnected, the flow shall be checked automatically. The coupling shall contain a positive locking device that will automatically lock the connection of the coupling halves. A spring force of 1.5 ± 0.5 pounds shall be provided to maintain the positive locking device in the locked position.

3.5.2 Type II coupling. The type II coupling shall contain no check valves. Gaseous oxygen shall flow through the coupling whether connected or disconnected.

* 3.5.3 Degreasing. All internal surfaces of the couplings shall be degreased by flushing with a cleaning compound, MIL-C-81302 or using a vapor degreaser in accordance with MIL-T-81533. Components shall be cleaned by immersing, scrubbing, or pressure spray with MIL-C-81302 cleaning compound or ultrasonics may be used in conjunction with vapor degreasing or MIL-C-81302 cleaning compound. Final rinse with MIL-C-81302 cleaning compound shall be with fresh, uncontaminated solvent. If vapor degreasing is used, the item shall be held in the vapor zone until all condensation has ceased prior to removal from vapor degreasing. After completion of the cleaning and when assembled, a General Electric Type H leak detector or equivalent Halide testing apparatus shall be used to determine the absence of the cleaning compound.

3.6 Performance.

3.6.1 Interchangeability.

3.6.1.1 Type I. When type I male and female coupling halves are connected respectively to qualified female and male coupling halves and tested as specified in 4.7.2.1, the coupling assembly shall conform to the following:

(a) Torque, type I	(3.6.2.1)
(b) Locking device, type I	(3.6.3)
(c) Leakage, type I	(3.6.4)
(d) Pressure drop, type I	(3.6.5)

3.6.1.2 Type II. When type II male and female coupling halves are connected respectively to qualified female and male coupling halves and tested as specified in 4.7.2.2, the maximum torque required to connect and disconnect the coupling halves shall not exceed 25 inch-pounds and the minimum disconnection torque shall be not less than 10 inch-pounds.

3.6.2 Torque.

3.6.2.1 Type I. The maximum torque required to connect and disconnect the coupling halves, when tested as specified in 4.7.3.1, shall not exceed 35 inch-pounds. The minimum torque required to disconnect the coupling halves shall be not less than 5 inch-pounds.

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3.6.2.2 Type II. The maximum torque required to connect and to disconnect the coupling halves, when tested as specified in 4.7.3.2, shall not exceed 25 inch-pounds. The minimum torque required to disconnect shall be not less than 10 inch-pounds.

3.6.3 Locking device, type I.

* 3.6.3.1 Without torque applied. When tested as specified in 4.7.4.1, the vertical force required to depress the positive locking device sufficiently to disengage it shall be 1.5 ± 0.5 pounds.

* 3.6.3.2 With torque applied. When tested as specified 4.7.4.2, the locking device shall not become disengaged.

3.6.4. Leakage, type I.

3.6.4 1 Coupling halves. The coupling leakage, when tested as specified in 4.7.5.1, shall not exceed 1 cubic centimeter (cc) per minute.

3.6.4.2 Coupling assembly. The coupling leakage, when tested as specified in 4.7.5.2, shall not exceed 1 cubic centimeter (cc) per minute.

3.6.4.3 Coupling assembly, with side load. The coupling leakage, when tested as specified in 4.7.5.3 shall not exceed 1 cubic centimeter (cc) per minute.

3.6.5 Pressure drop, type I. The pressure drop, when tested as specified in 4.7.6, shall be in accordance with Table I.

3.6.6 Proof pressure, type I. The coupling assemblies, when tested as specified in 4.7.7, shall not show any evidence of distortion, failure or malfunction.

3.6.7 Operational test.

* 3.6.7.1 Type I. The coupling assembly leakage, when tested as specified in 4.7.8.1, shall not exceed 1.0 cc/min and the disconnecting and connecting torque shall be as specified in 3.6.2.1.

3.6.7.2 Type II. The disconnecting and connecting torque, when tested as specified in 4.7.8.2, shall be as specified in 3.6.2.2.

3.6.8 Burst pressure (static), type I. The coupling, when tested as specified in 4.7.9, shall not rupture.

3.6.9 High temperature, types I and II. The type I or II coupling assembly, when tested as specified in 4.7.10.1 or 4.7.10.2 (as applicable), shall not show any evidence of mechanical or material failure.

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3.6.10 Vibration, types I and II. The type I or II coupling assembly, when tested as specified in 4.7.11.1 or 4.7.11.2 (as applicable), shall not become disconnected during the test and shall not show any evidence of mechanical or material failure.

3.6.11 Salt fog, types I and II. The type I or II coupling assembly, when tested as specified in 4.7.12.1 or 4.7.12.2 (as applicable), shall not show any evidence of corrosion or deterioration.

3.6.12 Reliability, types I and II. The type I or II coupling assembly shall have a specified mean cycles between failure (MCBF) of not less than 700 cycles at the 90 percent lower confidence limit. This is equivalent to a minimum acceptable MCBF of 466 cycles and to a reliability of .9986 for a one cycle mission.

3.6.13 Disassembly and examination, types I and II. When examined as specified in 4.7.14, the coupling assemblies shall not show evidence of excessive deterioration or wear as a result of testing which might affect the coupling performance.

* 3.7 Identification of product. Each coupling shall be permanently and legibly marked by stamping on the wrench flats or hex nuts in letters and numerals 1/16 inch (1.6 mm) or larger in height. The markings shall consist of the manufacturer's name or code identification, Government part number, and date of manufacture. On couplings with a sleeve lock, the above markings shall be stamped on the sleeve upper surface and shall also include the phrase "DEPRESS RING TO UNLOCK" and the circumferential score lines shown on MS22068.

3.8 Workmanship. The coupling assemblies shall be uniform in quality and shall be free from irregularities or defects which could adversely affect performance, reliability, or durability. The coupling assemblies shall be free of oil, grease, fuels, water, dust, dirt, or any other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The examination and testing of the couplings shall be classified as follows:

a. Qualification inspection (see 4.3)
b. First article inspection (see 4.4)
c. Quality conformance inspection (see 4.5)

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4.3 Qualification inspection. Qualification inspection shall consist of all the examinations and tests of this specification.

* 4.3.1 Qualification samples. Qualification samples shall consist of the following:

a. Four couplings of each type for which qualification is desired.

b. A Statement of Certification (with supporting test data) from the manufacturer that each type of coupling submitted conforms to the reliability requirements cited in paragraphs 3.6.12, 4.7.13 and 6.5.

c. Two sets of manufacturer's drawings for each type coupling submitted (see 6.5).

Samples shall be forwarded to the test facility set forth in the letter of authorization to submit samples (see 6.3). The samples shall be plainly identified by securely attached durable tags marked with the following information:

> Samples submitted by (name) on (date) for qualification inspection in accordance with the requirements of MIL-C-21049E under authorization (reference authorizing letter by code, serial number, and date) (see 6.3).

4.3.2 Retention. The retention of qualification shall consist of periodic verification to determined compliance of the qualified coupling with the requirements of this specification. The time and method of periodic verification shall be specified by the activity responsible for the Qualified Products List and shall be included in the Notice of Qualification letter.

* 4.4 First article inspection. First article inspection shall consist of the following examinations and tests, listed sequence mandatory:

Type I	Type II
Interchangeability	Interchangeability
Visual examination	Visual examination
Dimensions	Dimensions
Torque	Torque
Locking device	Operational
Leakage	Packaging
Pressure drop	
Proof pressure	
Operational	
Burst pressure	
Packaging	

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4.4.1 First article samples. Unless otherwise specified, as soon as practicable, after the award of the contract or order, the manufacturer shall submit two coupling assemblies or two coupling halves or each type specified in the contract or order. The samples shall be representative of the construction, workmanship, components, and materials to be used during production. When a manufacturer is in continuous production of these units from contract to contract, submission of further first article samples on the new contract may be waived at the discretion of the acquiring activity (see 6.2.1e). Approval of the first article samples or the waiving of first article inspection does not preclude the requirements of submitting to the quality conformance inspection. The first article inspection samples shall be furnished to the Government as directed by the contracting officer (see 6.2.1f).

* 4.4.1.1 First article information. Upon completion of the first article inspection, the Government activity responsible for conducting this inspection program (see 6.2.1f) shall report to the contracting officer the results of the inspection program, with appropriate recommendation.

* 4.4.1.2 Disposition of samples. Upon completion of the first article inspection program, disposition of samples shall be as follows:

a. Two approved samples shall be returned to the manufacturer for use in monitoring production.

b. Two samples shall not be returned due to their having been destroyed during the test program.

4.5 Quality conformance inspection. The sampling and inspection levels shall conform to MIL-STD-105. Quality conformance inspection shall consist of the following examinations and tests:

Visual examination
Dimensions
Torque
Leakage
Pressure drop
Proof pressure
Operational test
Packaging

4.5.1 Sampling.

4.5.1.1 Inspection lot.

4.5.1.1.1 Coupling assembly. An inspection lot size shall be expressed in units of coupling assemblies or coupling halves made essentially under the same conditions and from the same materials and components. The sample unit shall be one coupling assembly.

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4.5.1.1.2 Packaging. An inspection lot size shall be expressed in units of one fully prepared shipping container, containing couplings fully prepared for delivery, made from essentially the same materials and components. The sample unit shall be one shipping container, containing couplings fully prepared for delivery with the exception that it need not be sealed.

4.5.1.2 Sampling for tests and examinations of the coupling assemblies. The sample size, acceptance criteria, tests, and examinations required for the coupling assemblies shall be as specified in Table II.

4.6 Test conditions.

4.6.1 Oxygen. Oxygen used in testing the couplings shall be oxygen conforming to type I or type II, as applicable, of MIL-O-27210.

4.6.2 Temperature and pressure. Unless otherwise specified, tests shall be conducted at local ambient temperature and barometric pressure. The temperature and barometric pressure shall be recorded at the time of inspection. This information shall be available for computation of test data, where required, to normal temperature and pressure (NTP) conditions. NTP conditions are 29.92 inches of mercury and 70°F. Test instruments shall be calibrated or adjusted according to their required usage in conducting individual tests.

4.7 Inspection methods.

4.7.1 Visual examination.

4.7.1.1 Coupling assembly. Every coupling assembly shall be examined visually for defects to determine conformance to this specification. The classification of defects, Table III, shall be used to classify the defects found.

4.7.1.1.1 Dimensions. The couplings shall be checked dimensionally to determine coinformance to the dimensions specified in MS22068 and the automatic locking of the locking device when the type I coupling halves are connected. When measuring the travel of the lock sleeve, a constant angle shall be maintained between the diametrical plane of the lock sleeve and the longitudinal centerline of the coupling.

4.7.1.2 Packaging. Each of the fully prepared shipping containers, containing couplings selected as a sample unit from the lot shall be examined to determined that the packaging, packing and marking conform to this specification. The classification of detects, Table IV, shall be used to classify the defects found.

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4.7.2 Interchangeability.

4.7.2.1 Type I. The type I female coupling half shall be connected to each type I male coupling half that has been approved for listing on the Qualified Products List (QPL). Each coupling assembly shall then be subjected to the following tests:

Torque, type I (4.7.3.1)	Leakage, type I (4.7.5)
Locking device, type I (4.7.4)	Pressure drop, type I (4.7.6)

The above procedure shall then be repeated using the type I male coupling half with the female coupling halves that have been approved for listing on the QPL. Each combination coupling assembly shall pass the requirements specified in 3.6.1.1.

4.7.2.2 Type II. The type II female coupling half shall be connected to each type II male coupling half that has been approved for listing on the Qualified Products List (QPL). Each coupling assembly shall then be subjected to the torque type II test (4.7.3.2). The above procedure shall then be repeated using the type II male coupling half with the female coupling halves that have been approved for listing on the QPL. Each combination coupling assembly shall pass the requirements specified in 3.6.1.2.

4.7.3 Torque.

4.7.3.1 Type I. With the male and female halves connected and the outlet port capped, a pressure of 120 psig gaseous oxygen shall be applied to the inlet port. With this pressure maintained, and the locking device disengaged, the torque required to disconnect and connect the coupling halves shall be measured for five (5) cycles. One cycle shall consist of a disconnection and connection of the coupling halves. The torque required to disconnect and connect the coupling halves shall pass the requirements specified in 3.6.2.1.

4.7.3.2 Type II. The torque required to disconnect and connect the coupling halves shall be measured for five (5) cycles. One cycle shall consist of a disconnection and connection of the coupling halves. The torque required to disconnect and connect the coupling halves shall pass the requirements specified in 3.6.2.2.

4.7.4 Locking device, type I.

* 4.7.4.1 Without torque applied. With the coupling halves connected and the locking device engaged, the vertical force required to depress the positive locking device sufficiently to disengage it shall conform to the requirements specified in 3.6.3.1.

* 4.7.4.2 With torque applied. With the coupling halves connected and the locking device engaged, a torque of 35 inch pounds shall be applied to the coupling assembly in the direction of disconnection. The locking device shall pass the requirements specified in 3.6.3.2.

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* 4.7.5 Leakage, type I.

* 4.7.5.1 Coupling halves. The inlet end of the male and the outlet end of the halves shall each be connected to a gaseous oxygen source capable of providing a pressure up to 120 psig. With the coupling halves disconnected, pressures of 75 and 120 psig shall be applied to each coupling half. Each pressure shall be applied for a minimum of 3 minutes and leakage measured. The coupling shall pass the requirements specified in 3.6.4.1.

* 4.7.5.2 Coupling assembly. With the coupling halves connected, a pressure of 75 and 120 psig shall be applied to the inlet end of the male coupling with the outlet of the female coupling capped. Each pressure shall be applied for 3 minutes and leakage measured. The coupling shall pass the requirements specified in 3.6.4.2.

* 4.7.5.3 Coupling assembly, with side load. The coupling assembly shall be mounted to a test fixture in the manner shown in figure 1. The coupling assembly shall be in a vertical position with the male fitting end facing down and firmly connected to the test fixture. A pressure of 100 ± 10 psig shall be applied through the male fitting end. While maintaining this pressure, a 5 pound side load shall be applied to the upper edge of the capped outlet of the female coupling in the location and directions shown in figure 1. The load shall be maintained for a minimum of 1 minute in each direction and the leakage measured. The coupling shall pass the requirements specified in 3.6.4.3.

4.7.6 Pressure drop, type I. With the coupling halves connected, the coupling inlet shall be connected to a pressure source of 20 psig gaseous oxygen. With an inlet pressure of 20 psig, the pressure drop through the coupling assembly shall be determined for flow rates of 10, 60 and 120 liters per minute. The coupling assembly shall pass the requirements specified in 3.6.5.

4.7.7 Proof pressure, type I. With the coupling halves connected and the outlet port sealed, a pressure of 200 psig gaseous oxygen shall be applied to the inlet port of the coupling assembly. The coupling assembly shall pass the requirements specified in 3.6.6.

4.7.8 Operational test.

* 4.7.8.1 Type I. A liquid oxygen source of 60 to 90 psig shall be supplied to the inlet of the coupled assembly. The outlet shall have 6 to 12 inches of tubing connected to a manual valve. After starting a flow of liquid oxygen, submerge the cold coupling assembly in a bath of O-T-620 (trichloroethane) and flow for two minutes. After assuring that trapped air is released from the coupling, measure coupling leakage by the bubble collection method. Remove the assembly from the bath and immediately measure the torque required to disconnect and connect the coupling halves. Repeat the above procedure two additional times. The coupling assembly shall pass the requirements specified in 3.6.7.1. At the completion of this test, the coupling assembly shall be subjected to and pass the leakage, test specified in 4.7.5.

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4.7.8.2 Type II. The type II coupling assembly shall be connected to a liquid oxygen source and a flow of liquid oxygen produced through the coupling. The flow shall be stopped and the coupling halves immediately disconnected. The process of connecting the halves, producing a flow of liquid through the coupling, stopping the flow, and disconnecting the halves shall be repeated five times and the torque measured for each connection and disconnection. The coupling assembly shall pass the requirements specified in 3.6.7.2.

4.7.9 Burst pressure (static), type I. With the outlet port of the coupling sealed, a gaseous oxygen pressure of 350 psig shall be applied to the inlet port of the coupling and maintained for 15 seconds. The coupling shall pass the requirements specified in 3.6.8.

4.7.10 High temperature.

4.7.10.1 Type I. The coupling assembly shall be placed in a chamber having an internal temperature of 260 ± 5°F (127 ± 3°C) for a minimum period of 1 hour. After this exposure, the coupling shall be removed from the chamber and immediately subjected to and pass the torque, type I (4.7.3.1) and leakage, type I (4.7.5) tests. The coupling assembly shall pass the requirements specified in 3.6.9.

4.7.10.2 Type II. The high temperature test shall be the same as for type I except that the coupling assembly shall be subjected only to the torque, type II test (4.7.3.2) after temperature conditioning. The coupling assembly shall pass the requirements specified in 3.6.9.

4.7.11 Vibration. The coupling assembly shall be subjected to a random vibration test in accordance with MIL-STD-810, Method 514.3, Procedure I, Category 5, except the test conditions shall be in accordance with 4.6.2. W_o shall be 0.30 G^2/Hz and the test duration shall be one hour in each axis.

4.7.11.1 Type I. The coupling assembly shall be tested, as specified in 4.7.11, with the coupling assembly outlet capped and a pressure of 120 to 150 psig applied to the coupling assembly inlet for approximately one-half the test time in each axis. The remaining test time shall be performed with the coupling assembly outlet open and a pressure of 60 to 90 psig applied to the coupling assembly inlet with a cryogenic gas of a temperature to establish an internal coupling temperature or -250°F or colder. During this vibration, a torque of 35 inch-pounds shall be intermittently applied in a disconnect direction at intervals of five minutes. The coupling assembly shall pass the requirements specified in 3.6.10. Upon completion of vibration, the coupling assembly shall be subjected to, and pass the torque, type I (4.7.3.1) and the leakage, type I (4.7.5) tests.

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4.7.11.2 Type II. The coupling assembly shall be tested, as specified in 4.7.11, with no internal pressure for approximately one-half the test time in each axis. The remaining test time shall be with a flow of cryogenic gas of a temperature to establish an internal coupling temperature of -250°F or colder. The coupling assembly shall pass the requirements specified in 3.6.10. Upon completion of the vibration test, the coupling assembly shall be subjected to, and pass the torque, type II test (4.7.3.2).

4.7.12 Salt fog.

4.7.12.1 Type I. The coupling assembly shall be subjected to a salt fog test in accordance with MIL-STD-810, Method 509.2, Procedure I, except the salt fog exposure period shall be 168 hours. The coupling assembly shall pass the requirements specified in 3.6.11. After completing the salt fog test, the coupling assembly shall be subjected to and pass the following tests:

Torque, type I (4.7.3.1)
Locking device, type I (4.7.4)
Leakage, type I (4.7.5)

4.7.12.2 Type II. The salt fog test shall be the same as for type I, except the coupling assembly shall be subjected only to the torque, type II test (4.7.3.2) after the salt fog test. The coupling assembly shall pass the requirements specified in 3.6.11.

4.7.13 Reliability.

4.7.13.1 Type I. The coupling assembly shall be subjected to cycles of disconnections and connections. A cycle shall consist of a disconnection (coupling halves separated) and a connection with the coupling outlet port sealed and 120 psig gaseous oxygen pressure applied to the coupling inlet port. The coupling assembly shall be subjected to the torque, type I (4.7.3.1), locking device, type I (4.7.4) and the leakage, type I (4.7.5) tests every 700 cycles during the reliability test to determine if a failure has occurred. A coupling assembly failing any of these tests shall have failed the reliability test. In addition, any failure of the coupling to mate properly or fracture of components shall constitute a test failure. No parts shall be replaced during the reliability testing as preventive maintenance. Acceptance shall be based on the accept-reject criteria of test plan I of MIL-STD-781. Recording, data handling, and reporting procedures shall be in accordance with MIL-STD-781.

4.7.13.2 Type II. Reliability testing shall be the same as for type I except that the coupling assembly shall be unpressurized during the test and shall be subjected only to the torque, type II test (4.7.3.2) every 700 cycles.

4.7.14 Disassembly and examination types I and II. At the conclusion of all tests specified herein or after any test in which the coupling failure indicates faulty construction, the coupling assembly shall be disassembled and examined. The coupling assembly shall pass the requirements specified in 3.6.13.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A, B or C, as specified by the acquiring activity (see 6.2.1g).

5.1.1 Levels A & B. Each coupling shall be preserved and packaged in accordance with method ICI of MIL-P-116 without the use of contact preservative. Cleaning shall be accomplished in accordance with industry practice for oxygen service. Petroleum derivative solvents shall not be used. The bagged coupling shall be cushioned with material conforming to PPP-C-843, minimum thickness of one inch and then enclosed within a heat sealed barrier bag fabricated from material conforming to MIL-B-121, Grade A. Prior to sealing all excess air must be exhausted. Each coupling shall then be packaged within an individual close-fitting box conforming to PPP-B-566, PPP-B-665 or PPP-B-676. The closure shall be accomplished in accordance with the procedures specified in the appendix of the applicable specification.

5.1.2 Level C. Couplings shall be packaged in a manner that will prevent physical damage or environmental deterioration during transit from the shipping point to the original consignee.

5.2 Packing. Packing shall be Level A, B or C as specified by the acquiring activity (see 6.2.1g). Insofar as practicable shipping containers shall effect a close fit, contain identical quantities and be of uniform dimensional configuration.

5.2.1 Level A. Couplings packaged as specified in 5.1.1 shall be packed for shipment in fiberboard shipping container conforming to PPP-B-636, class-weather resistant Grade V3S/V3C. The gross weight of each packed container shall not exceed 70 pounds. The closure, sealing and reinforcing (banding) of the containrs shall be accomplished in accordance with procedure specified in the appendix of the container specification.

5.2.2 Level B. Couplings packaged as specified in 5.1.1 shall be packed for shipment as specified in 5.2.1 except that the fiberboard shipping containers conforming to PPP-B-636, shall be Class-Domestic.

5.2.3 Level C. Couplings packaged as specified, shall be packed for shipment in a manner that will afford protection against damage during transit from the shipping point to the original consignee. Containers shall be in accordance with Uniform Freight Classification Rules or regulations or other carriers applicable to the mode of transportation utilized.

5.3 Marking. Each unit and shipping container shall be marked in accordance with MIL-STD-129. In addition, the following precautionary marking shall be required on each unit package and shipping container (see 6.2.1i):

CAUTION

DO NOT ALLOW CONTAMINANTS OF ANY KIND TO BE USED ON OR ABOUT OXYGEN COUPLING PACKAGES.

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6. NOTES

6.1 Intended use. The couplings are intended for use in aircraft liquid oxygen systems to uncouple the line without draining the system and without undue loss of liquid oxygen.

6.2 Ordering data.

* 6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification (including amendments).

b. Types of coupling assembly or coupling half required (see 1.2).

c. Quantity required.

d. Applicable National Stock Number (NSN).

e. Whether first article inspection if waived (see 4.4.1).

f. Name and address of the first article inspection test facility (see 4.4.1) and the name and address of the Government activity responsible for conducting the first article inspection program (see 4.4.1.1 and 6.4).

g. Selection of applicable levels of preservation, packaging, and packing (see 5.1 and 5.2).

h. Applicable methods of cleaning and preservation.

i. Whether any special markings are required (see 5.3).

* 6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9(n)(2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraph:

Paragraph	Data Requirement	Applicable DID
4.4.1.1	First Article Inspection Reports	DI-T-5329
4.7.13	Reliability Inspection Reports	DI-T-5329

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6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Commander, Naval Air Systems Command, Department of the Navy, Washington, DC 20360; however, authorization for qualification of products shall be obtained from the Commanding Officer, Naval Air Development Center, (Code 6031) Warminster, PA 18974. Prior to submission of the samples for qualification inspection, the manufacturer shall submit a request to the Naval Air Development Center (Code 6031) indicating a date on which the samples can be forwarded and also request an authorization number to accompany the samples.

* 6.4 First article. When a first article is required, the item shall be tested and should be a first article sample. The first article sample shall not be one of the initial production items. The manufacturer should not enter into production until he has first article approval. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for examinations, test and approval of the first article.

* 6.5 Documentation. When requested, the manufacturer shall submit engineering drawings and inspection reports in accordance with SD-6, Provisions Governing Qualifications.

6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Navy - AS
Air Force - 99

Preparing activity:
Navy - AS
(Project No. 1660-0516)

Review Activities:
Air Force - 11, 71

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TABLE I. Pressure drop.

Maximum pressure drop (inches - Hg)	Flow (liters per minute)
0.20	10
3.5	60
15.0	120

TABLE II. Sample size, acceptance criteria, tests and examination of the coupling assemblies.

INSPECTION	METHOD	SAMPLE SIZE	ACCEPTANCE CRITERIA
Visual examination (See classification of defects)	4.7.1.1	a. Every coupling for critical defects. b. Inspection Level II for minor defects 1/	a. Reject all units with any critical defects. b. An acceptable quality level of 2.5 defects per hundred units for minor defects
Dimensions	4.7.1.1.1	Inspection level S-2 1/	Acceptance number zero, rejection number 1
Torque	4.7.3	Every coupling	Reject all defective units
Leakage, Type I, coupling halves	4.7.5.1	Every coupling	Reject all defective units
Leakage, Type I, coupling assy.	4.7.5.2	Every coupling	Reject all defective units
Leakage, Type I, coupling assy. with side load	4.7 5.3	Inspection level S-2 1/	Acceptance number zero, rejection number 1
Pressure drop type I	4.7.6	Inspection Level S-2 1/	Acceptance number zero, rejection number 1
Proof pressure type I	4.7.7	Inspection Level S-2 1/	Acceptance number zero, rejection number 1
Operational test	4.7.8	Inspection Level S-2 1/	Acceptance number zero, rejection number 1
Packaging	4.7.1.2	Inspection Level S-2 1/	Total acceptable quality level of 4.0 percent defective

1/ The sample size shall be based only on the applicable sample size code letter corresponding to the specified inspection level of MIL-STD-105.

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TABLE III. Classification of defects for visual examination of the coupling assemblies.

Critical	Minor
1. Material imperfection or not as specified.	201. Marking missing, insufficient, incorrect, illegible, or not permanent.
2. Surface unclean, rough, misaligned, containing cracks, nicks, or other flaws.	
3. Any component missing, malformed, fractured, or otherwise damaged.	
4. Any component loose or not securely retained.	
5. Incorrect assembling or improper positioning of components.	
6. Any functioning part that works with difficulty.	
7. Locking device of type I does not go in lock detents when coupling halves are connected.	
8. Internal passage through the type II assembly is obstructed.	
9. Dimensions of threads not as specified.	
10. Faulty workmanship.	

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TABLE IV. List of defects for packaging.

Item	Defects
Exterior and interior markings	Missing, incorrect, incomplete, illegible; or improper size, loction, sequence, or method or application; markings not the same on the interior containers.
Packaging and packing materials	Any nonconforming component; any component missing, damaged, or otherwise defective.
Workmanship	Inadequate application of the components such as incomplete closure of the unit package, intermediate package, container flaps, loose strappings, etc.; bulging or distortion of the containers.
Exterior and interior weight or content	Number per container is more or less than required; gross or net weight exceeds the requirements; more than one type in the same container.

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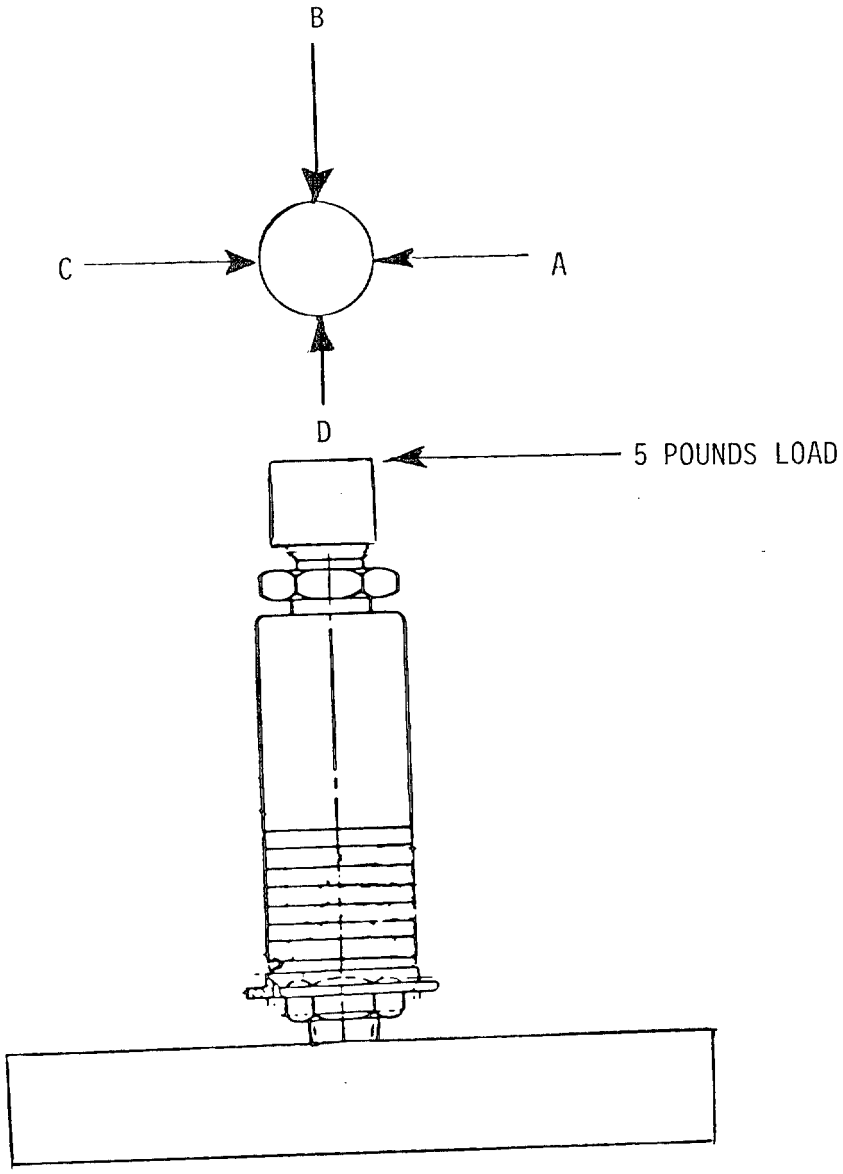


Figure 1. Leakage test with side load.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-21049E	Coupling Assemblies, Quick Disconnect, Aircraft LOX Systems

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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